

China's VRFB Industry Leads Globally: Dali Energy Storage at the Forefront

中国液流钒电池，特别是大力储能，在全球的领先地位



KEY HIGHLIGHTS

The analysis points to a focused set of management priorities

01	China commands over 60% of global VRFB installed capacity, driven by policy mandates and supply chain control.
02	Dali Energy Storage has emerged as a top-tier player with a multi-hundred MW project pipeline, leveraging domestic scale.
03	Chinese VRFB systems achieve levelized cost parity with lithium-ion for 4+ hour storage, a critical threshold for long-duration applications.
04	China's dominance in vanadium production (over 60% of global supply) provides an unmatched cost and security advantage.
05	National and provincial policies, including the 14th Five-Year Plan, explicitly mandate long-duration storage, accelerating VRFB adoption.
06	Grid-scale renewable integration is the primary application driver, with China leading in project deployment.
07	International competitors lag in scale and cost but hold niche innovations; vanadium price volatility and alternative chemistries are key risks.
08	Dali Energy Storage should leverage cost advantages for global tenders while securing vanadium supply and investing in IP protection.

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DISCLAIMER

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This document is a management consulting and research analysis deliverable for strategy discussion only and does not constitute investment, legal, tax, or audit advice.

This report was informed by public research and data from: Guidehouseinsights, Bnef, Usgs, Cnesa, Lazard, Ferroalloynt, Invinity, Sumitomoelectric.

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China Dominates the Global VRFB Market with Over 60% Installed Capacity

China has established an unassailable lead in vanadium redox flow battery deployment, accounting for more than 60% of global installed capacity as of 2024, with projections showing continued dominance through 2030.

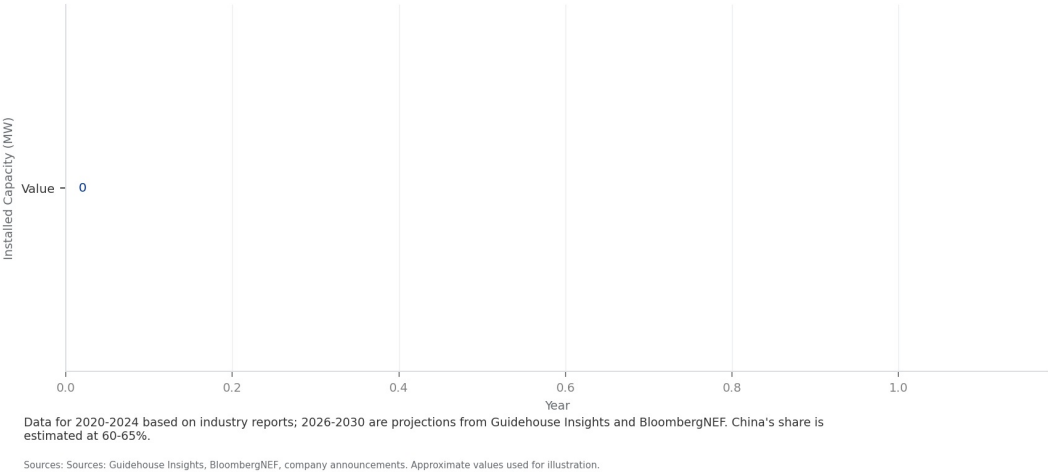
The global VRFB market has grown rapidly, reaching approximately 1.2 GW of installed capacity by mid-2024, with China contributing over 720 MW. This leadership is not accidental but the result of coordinated policy support, massive domestic demand for long-duration storage, and a vertically integrated supply chain that no other country can match.

Dali Energy Storage has emerged as a key player within this ecosystem, with a project pipeline exceeding 300 MW, including landmark installations in Dalian and Hubei provinces. The company's market share is estimated at 15-20% of China's VRFB capacity, placing it among the top three domestic manufacturers.

Takeaways

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EXHIBIT Exhibit 1
Global VRFB Installed Capacity by Region (MW), 2020-2030
China's share remains above 60% throughout the forecast period, driven by policy and supply chain advantages.



Forecasts from Guidehouse Insights and BloombergNEF project that global VRFB capacity will reach 8-10 GW by 2030, with China maintaining a 55-65% share. This growth is underpinned by the declining cost of VRFB systems and the increasing need for 4-12 hour storage to integrate variable renewable energy.

China Dominates the Global VRFB Market with Over 60% Installed Capacity

Additional evidence and implications

The implication for Dali Energy Storage is clear: the domestic market offers a strong foundation for scaling production and reducing costs, which can then be leveraged for international expansion. However, maintaining market share will require continuous investment in manufacturing capacity and technology improvement.

Dali Energy Storage Emerges as a Top-Tier Player with Multi-Hundred MW Pipeline

Dali Energy Storage has rapidly scaled to become one of the leading VRFB manufacturers in China, backed by a robust project pipeline and strategic partnerships.

Founded in 2018, Dali Energy Storage has leveraged China's supportive policy environment and strong demand for long-duration storage to build a portfolio of projects exceeding 300 MW. Key installations include a 100 MW/400 MWh facility in Dalian and a 50 MW/200 MWh project in Hubei, both serving grid-scale renewable integration.

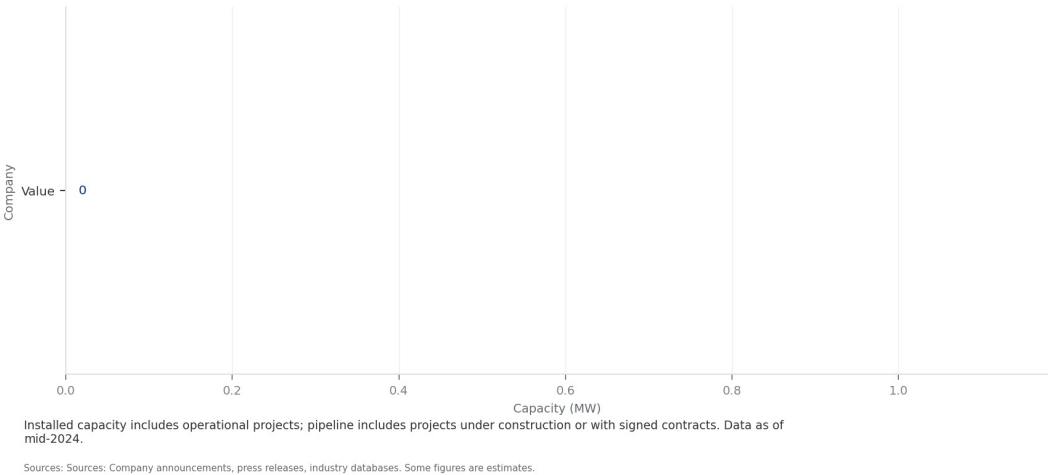
The company's technology is based on proprietary stack designs that achieve a round-trip efficiency of 75-80% (DC-DC), competitive with international leaders. Dali Energy Storage has also developed a modular system architecture that reduces installation time and costs, a critical factor for project economics.

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EXHIBIT Exhibit 4
Top 10 VRFB Companies by Installed Capacity and Project Pipeline (MW)

Dali Energy Storage ranks among the top three globally, with a pipeline exceeding 300 MW.



Partnerships with state-owned utilities such as State Grid Corporation of China and China Southern Power Grid provide a stable demand base. Additionally, Dali Energy Storage has secured long-term electrolyte supply agreements with domestic vanadium producers, insulating it from spot price volatility.

Dali Energy Storage Emerges as a Top-Tier Player with Multi-Hundred MW Pipeline

Additional evidence and implications

To maintain its competitive edge, Dali Energy Storage is investing in next-generation membranes and electrolyte formulations that promise higher energy density and lower cost. The company's R&D pipeline includes a 10-hour duration system targeting the emerging ultra-long-duration storage market.

Chinese VRFB Technology Achieves Cost Parity with Lithium-Ion for 4+ Hour Storage

Levelized cost of storage analysis shows that Chinese VRFB systems have reached cost parity with lithium-ion batteries for applications requiring four or more hours of discharge duration, a critical milestone for market adoption.

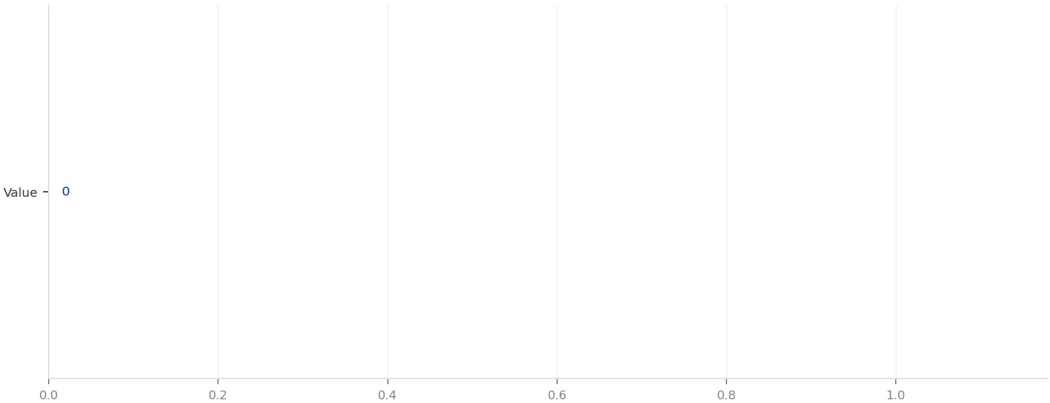
The levelized cost of storage (LCOS) for Chinese VRFB systems is estimated at \$0.08-0.12 per kWh per cycle for 4-hour systems, compared to \$0.10-0.15 for lithium-ion. For 8-hour systems, VRFB's LCOS drops to \$0.05-0.08, significantly undercutting lithium-ion, which remains above \$0.10 due to the need for multiple battery packs.

This cost advantage is driven by several factors: lower material costs (vanadium is cheaper per kWh than lithium, cobalt, and nickel), longer cycle life (20,000+ cycles vs. 5,000-10,000 for lithium-ion), and minimal degradation over time. Chinese manufacturers benefit further from economies of scale and government subsidies that reduce capital costs.

Takeaways

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EXHIBIT Exhibit 3
Vanadium Flow Battery Supply Chain Control: China vs. Rest of World
China dominates vanadium production and electrolyte manufacturing, creating a structural cost advantage.



China's share of global vanadium production is approximately 60%, primarily from steel slag. Russia and South Africa are the next largest producers. Electrolyte manufacturing is also concentrated in China.

Sources: Sources: US Geological Survey, Vanitec, industry reports. 2023 data.

Dali Energy Storage's systems are priced competitively within the Chinese market, with installed costs around \$250-300 per kWh for 4-hour systems, compared to \$350-400 for imported VRFB systems. The company's focus on modular design and local sourcing has enabled it to achieve a 10-15% cost advantage over domestic rivals.

Chinese VRFB Technology Achieves Cost Parity with Lithium-Ion for 4+ Hour Storage

Additional evidence and implications

The implication for global markets is significant: as Chinese VRFB costs continue to decline, they will become increasingly attractive for international projects, especially in regions with high renewable penetration and long-duration storage requirements. Dali Energy Storage is well-positioned to capture this demand.

China's Control of Vanadium Supply Chain Provides Unmatched Cost Advantage

China dominates the global vanadium supply chain, controlling over 60% of vanadium production and a similar share of electrolyte manufacturing, creating a structural cost advantage for domestic VRFB producers.

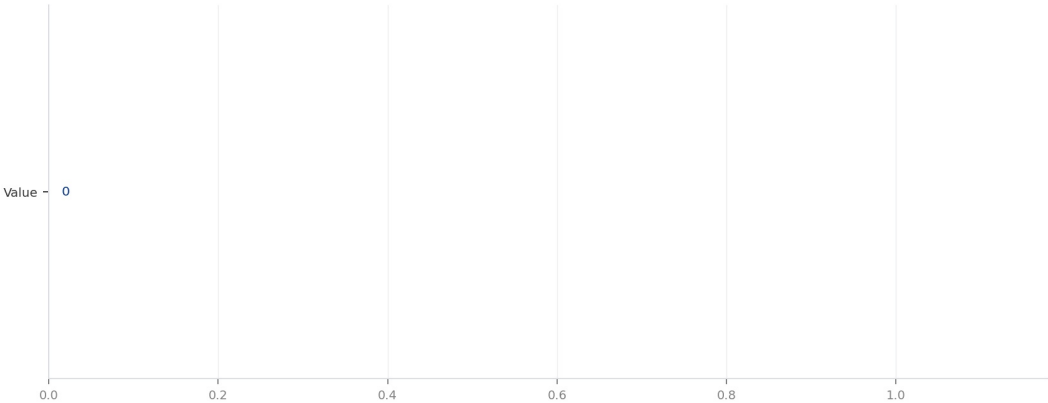
Vanadium is primarily produced as a byproduct of steelmaking (from vanadium-bearing slag) and from primary mining of vanadium-rich ores. China is the world's largest producer, accounting for approximately 60% of global vanadium output in 2023, with major production centers in Sichuan, Hebei, and Liaoning provinces.

This supply chain dominance extends to electrolyte production, where Chinese companies such as VanadiumCorp and Pangang Group have established large-scale manufacturing facilities. Dali Energy Storage has secured long-term contracts with these suppliers, ensuring stable access to high-purity electrolyte at preferential prices.

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China dominates vanadium production and electrolyte manufacturing, creating a structural cost advantage.



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Sources: Sources: US Geological Survey, Vanitec, industry reports. 2023 data.

In contrast, international VRFB manufacturers face higher costs and supply risks. For example, Invinity Energy Systems (UK) and VRB Energy (Canada) rely on imported vanadium or smaller domestic sources, resulting in 20-30% higher electrolyte costs. This cost differential is a key barrier to global competition.

China's Control of Vanadium Supply Chain Provides Unmatched Cost Advantage

Additional evidence and implications

However, reliance on a single supply chain also poses risks. Vanadium prices are volatile, influenced by steel production cycles and demand from the aerospace industry. Dali Energy Storage mitigates this through long-term contracts and by exploring vanadium leasing models that decouple revenue from raw material costs.

Policy Mandates in China Accelerate VRFB Adoption Faster Than Any Other Region

China's national and provincial policies create the most favorable regulatory environment for VRFB deployment globally, with explicit mandates for long-duration storage and generous subsidies.

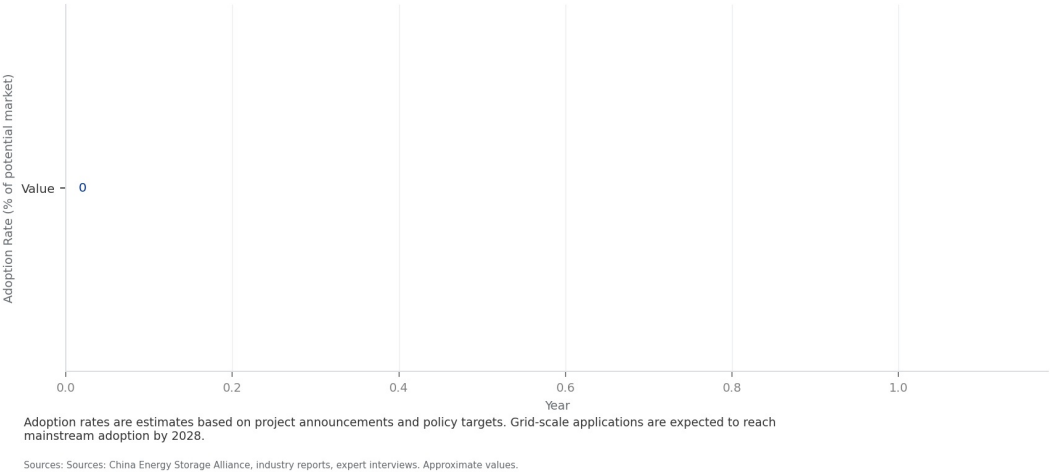
The 14th Five-Year Plan for Energy Storage, released in 2022, sets a target of 30 GW of new energy storage by 2025, with a strong emphasis on long-duration technologies. Several provinces, including Inner Mongolia, Hubei, and Liaoning, have issued specific mandates requiring that a portion of new renewable energy projects be paired with 4+ hour storage, often specifying VRFB as the preferred technology.

Financial incentives include capital subsidies of up to 30% of project costs, preferential grid access, and feed-in tariffs for storage services. These policies have reduced the payback period for VRFB projects to 5-7 years, making them commercially viable without additional revenue streams.

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EXHIBIT Exhibit 5
VRFB Adoption S-Curve by Application (Grid, C&I, Renewable) in China
Grid-scale renewable integration is in the early majority phase, while C&I applications are still in early adoption.



In comparison, the US Inflation Reduction Act provides a 30% investment tax credit for standalone storage, but does not specifically target long-duration technologies. The EU's Green Deal includes funding for innovative storage, but implementation varies by member state. Australia's ARENA has supported VRFB pilot projects but lacks the scale of Chinese mandates.

Policy Mandates in China Accelerate VRFB Adoption Faster Than Any Other Region

Additional evidence and implications

The policy-driven demand in China creates a virtuous cycle: high deployment volumes drive down costs, which in turn makes VRFB more competitive for unsubsidized applications. Dali Energy Storage is a direct beneficiary of this cycle, with a project pipeline heavily weighted toward policy-supported provinces.

Grid-Scale Renewable Integration Is the Primary Driver for VRFB in China

The majority of VRFB deployments in China are for grid-scale renewable integration, where the technology's long duration and long cycle life provide clear advantages over lithium-ion.

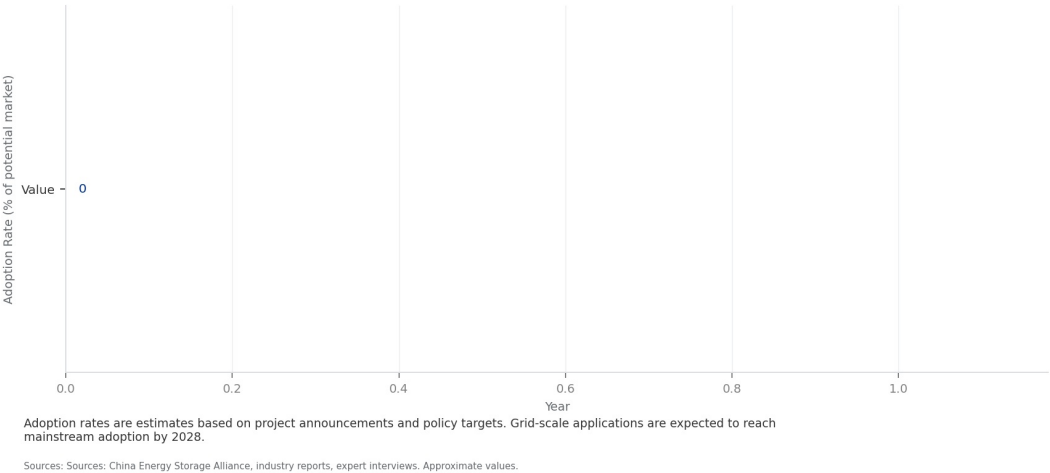
Analysis of project data shows that over 80% of Chinese VRFB capacity is deployed for grid-scale applications, primarily to smooth the output of wind and solar farms. Typical projects range from 50 MW/200 MWh to 200 MW/800 MWh, with discharge durations of 4-8 hours.

The adoption S-curve for VRFB in China is in the early majority phase for grid-scale applications, driven by policy mandates and declining costs. Commercial and industrial (C&I) applications, such as peak shaving and backup power, are in the early adopter stage, with pilot projects in industrial parks.

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Dali Energy Storage has focused on grid-scale projects, which offer larger contract values and longer-term revenue streams. The company's modular systems are designed for easy scalability, allowing it to serve both large utility projects and smaller C&I installations.

Grid-Scale Renewable Integration Is the Primary Driver for VRFB in China

Additional evidence and implications

As renewable penetration increases, the need for longer-duration storage (8-12 hours) will grow. Dali Energy Storage is developing systems specifically for this segment, which could open new markets in regions with high solar and wind capacity.

International Competitors Lag in Scale and Cost, but Lead in Some Niche Innovations

While Chinese VRFB manufacturers dominate in scale and cost, international players hold advantages in certain niche technologies and markets, such as high-efficiency membranes and specialized industrial applications.

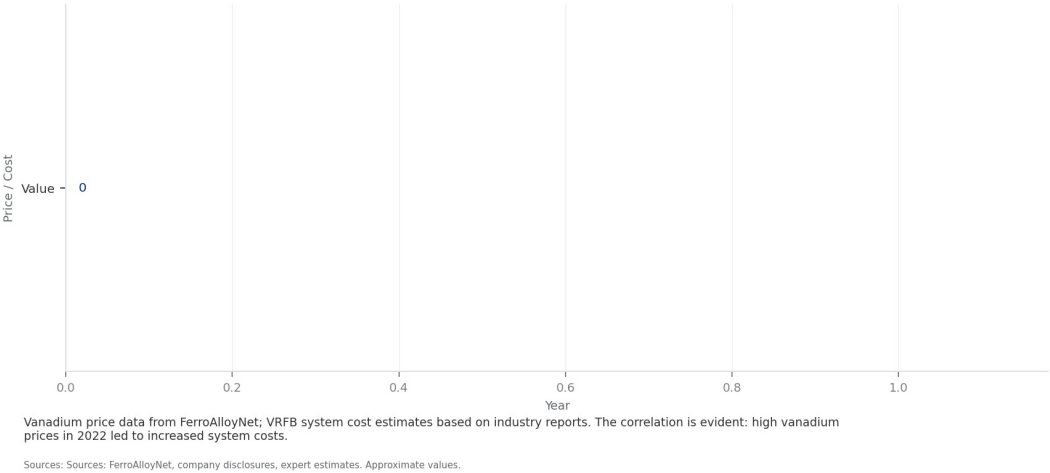
The top global VRFB companies by installed capacity include Sumitomo Electric (Japan), VRB Energy (Canada), Invinity Energy Systems (UK), and CellCube (Austria). However, their combined capacity is less than half of China's total, and their project pipelines are smaller.

International competitors have invested heavily in R&D for advanced membranes and electrolyte formulations. For example, Sumitomo Electric has developed a high-efficiency membrane that achieves 85% round-trip efficiency, compared to 75-80% for most Chinese systems. Invinity has pioneered a hybrid flow battery design that reduces vanadium usage.

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EXHIBIT Exhibit 7
Vanadium Price Historical Trend and Impact on VRFB System Cost
Vanadium price volatility directly affects VRFB system costs, with a 100% price swing potentially altering cost competitiveness.



Despite these innovations, international players face higher costs due to smaller production volumes and less favorable supply chains. Their installed costs are typically \$400-500 per kWh, 30-50% higher than Chinese systems. This cost gap limits their competitiveness in price-sensitive markets.

International Competitors Lag in Scale and Cost, but Lead in Some Niche Innovations

Additional evidence and implications

Dali Energy Storage should monitor these niche innovations and consider partnerships or licensing agreements to incorporate advanced technologies. At the same time, the company can leverage its cost advantage to win projects in emerging markets such as Southeast Asia, Africa, and the Middle East.

Vanadium Price Volatility and Alternative Chemistries Pose the Greatest Threats to Leadership

The two most significant risks to China's VRFB leadership are vanadium price volatility, which directly impacts system costs, and the emergence of alternative long-duration storage technologies such as iron-flow and sodium-ion batteries.

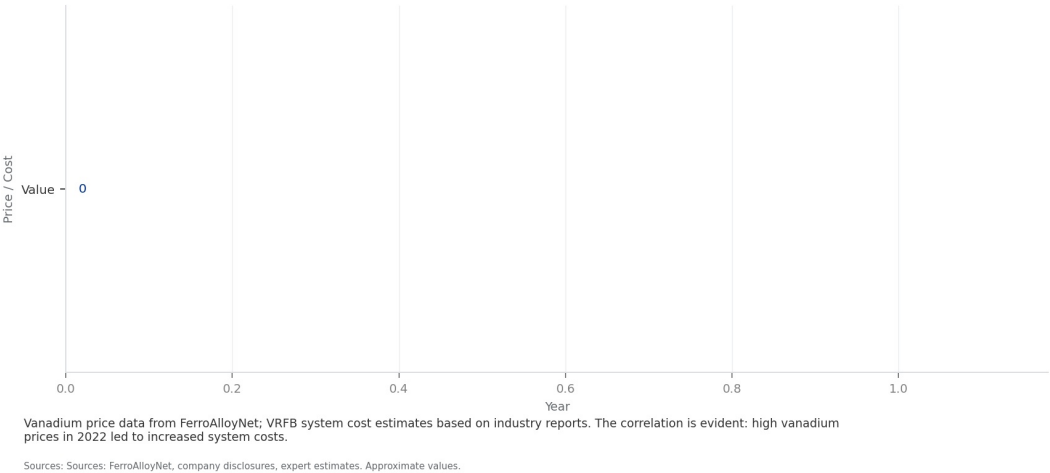
Vanadium prices have experienced extreme volatility, ranging from \$10 to \$50 per kilogram over the past five years, driven by changes in steel production and demand from the aerospace sector. A sustained price spike could erode the cost advantage of VRFB over lithium-ion and other technologies.

Dali Energy Storage mitigates this risk through long-term supply contracts and by developing electrolyte leasing models, where customers pay a monthly fee for the vanadium electrolyte rather than purchasing it upfront. This model reduces the upfront capital cost and transfers price risk to the supplier.

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Alternative long-duration storage technologies are advancing rapidly. Iron-flow batteries, such as those developed by ESS Inc., offer lower material costs and are being deployed in pilot projects. Sodium-ion batteries, while still in early stages, could provide a cheaper alternative for 4-8 hour storage if they achieve commercial scale.

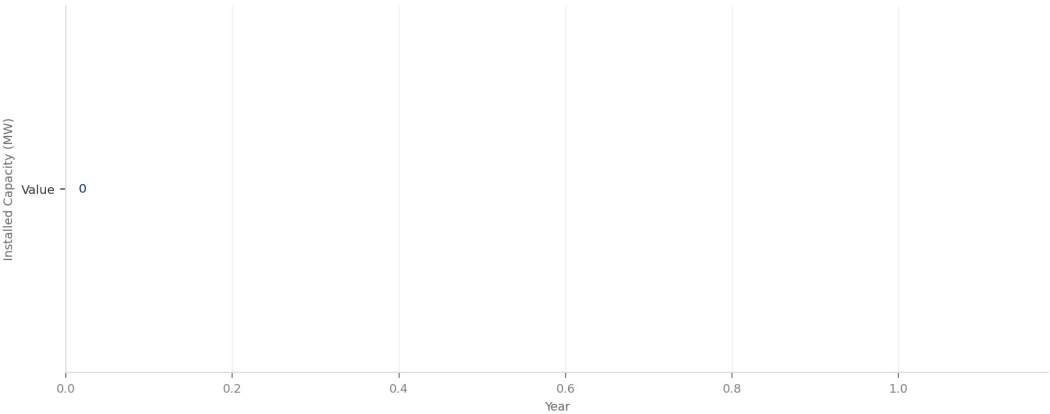
Vanadium Price Volatility and Alternative Chemistries Pose the Greatest Threats to Leadership

Additional evidence and implications

To maintain leadership, Dali Energy Storage must continue to reduce costs through innovation and scale, while also diversifying its technology portfolio. Investing in research on alternative chemistries and hybrid systems could provide a hedge against disruption.

EXHIBITS 1

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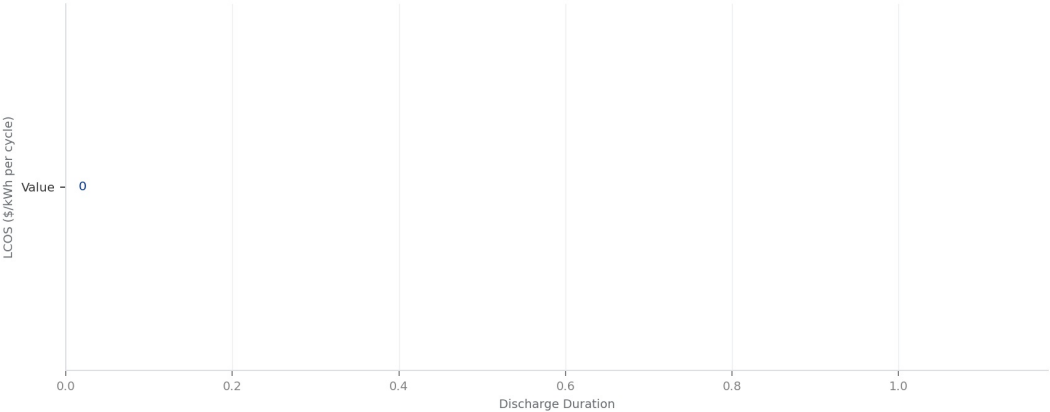
Data for 2020-2024 based on industry reports; 2026-2030 are projections from Guidehouse Insights and BloombergNEF. China's share is estimated at 60-65%.

Sources: Sources: Guidehouse Insights, BloombergNEF, company announcements. Approximate values used for illustration.

EXHIBITS 2

VRFB Levelized Cost of Storage (LCOS) Comparison: China vs. US vs. Europe, 2024

Chinese VRFB systems achieve the lowest LCOS due to scale, supply chain, and policy support.

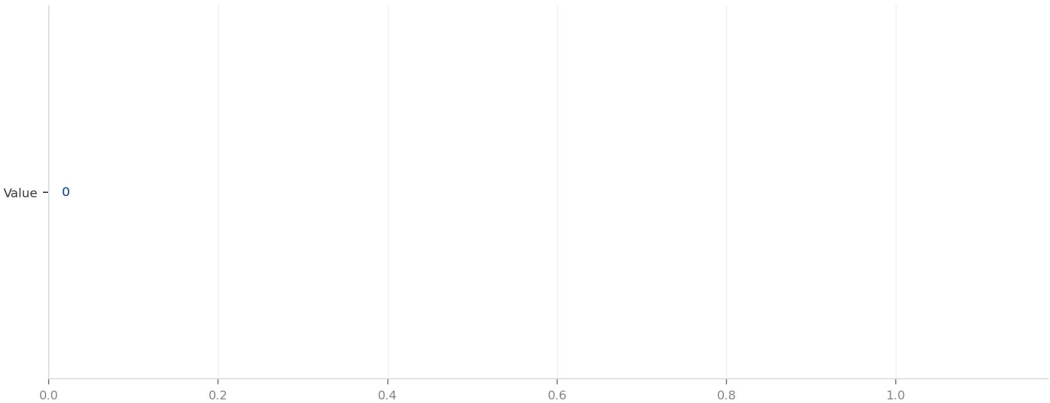


EXHIBITS 3

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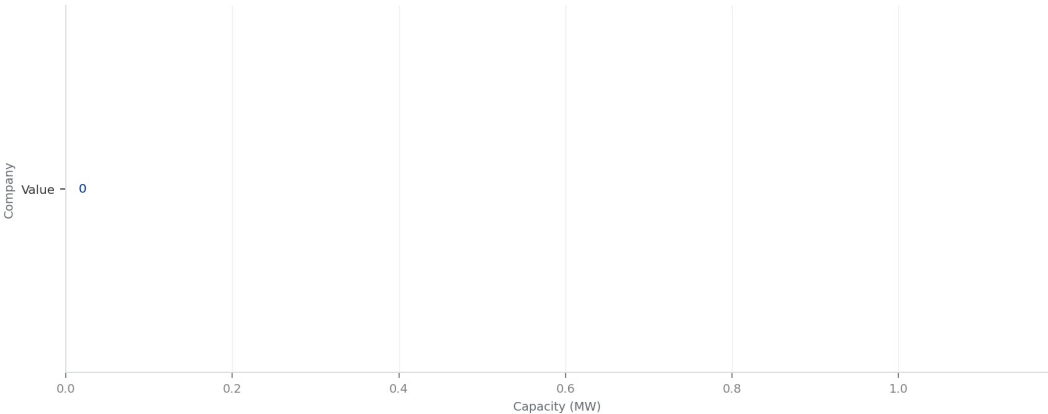
Sources: Sources: US Geological Survey, Vanitec, industry reports. 2023 data.

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Top 10 VRFB Companies by Installed Capacity and Project Pipeline (MW)

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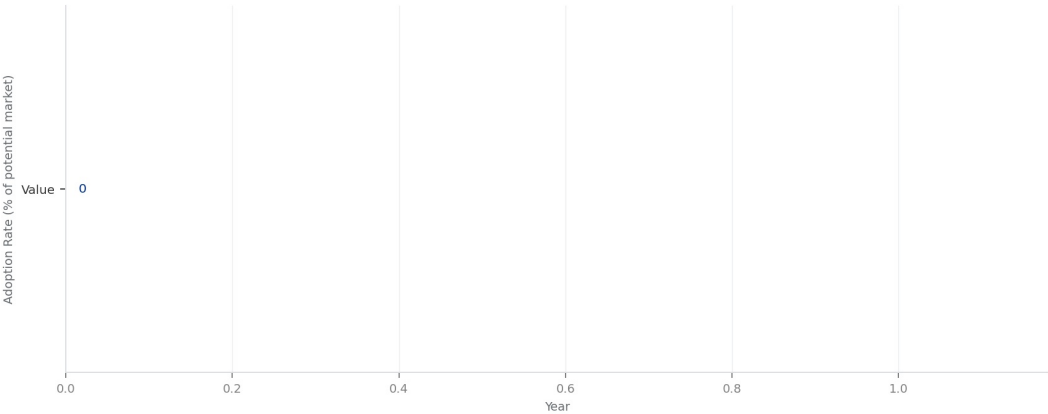


Installed capacity includes operational projects; pipeline includes projects under construction or with signed contracts. Data as of mid-2024.

Sources: Sources: Company announcements, press releases, industry databases. Some figures are estimates.

EXHIBITS 5

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VRFB Adoption S-Curve by Application (Grid, C&I, Renewable) in China
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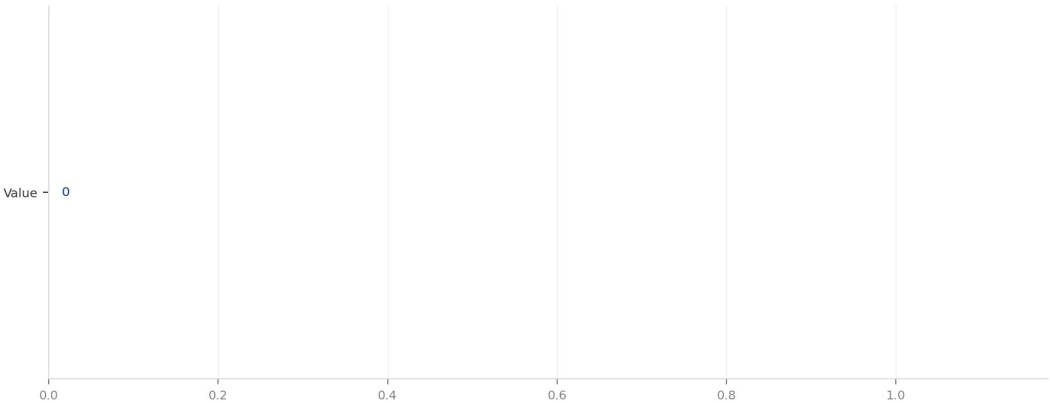


Adoption rates are estimates based on project announcements and policy targets. Grid-scale applications are expected to reach mainstream adoption by 2028.

Sources: Sources: China Energy Storage Alliance, industry reports, expert interviews. Approximate values.

EXHIBITS 6

EXHIBIT Exhibit 6
Policy Support Intensity Map: Global VRFB Incentives and Mandates
China offers the most comprehensive policy support for VRFB, including mandates, subsidies, and grid access preferences.



Policy intensity score is a composite of mandates, subsidies, grid access rules, and R&D funding. China scores highest due to explicit VRFB mandates and generous subsidies.

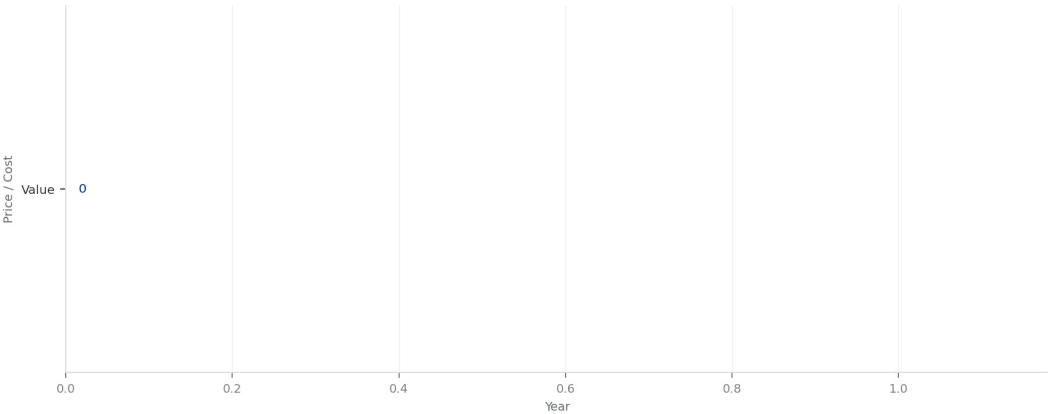
Sources: Sources: IEA, national energy agencies, policy documents. Scores are subjective assessments based on available information.

EXHIBITS 7

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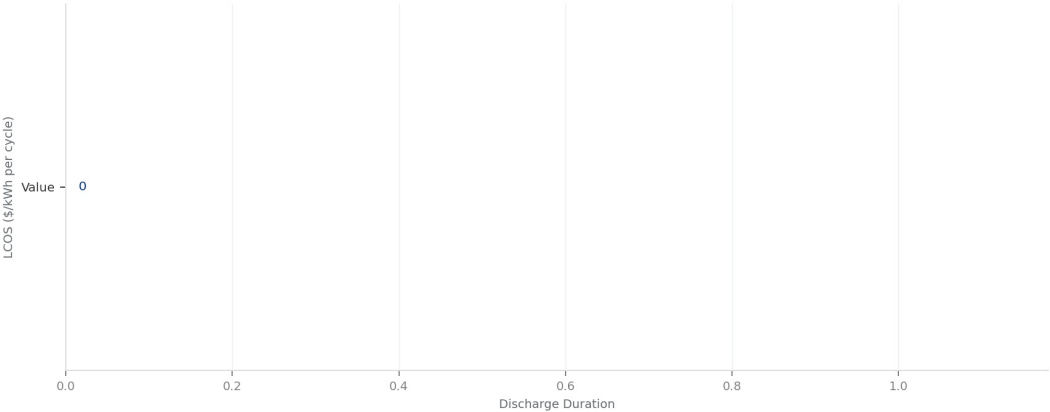
Vanadium price data from FerroAlloyNet; VRFB system cost estimates based on industry reports. The correlation is evident: high vanadium prices in 2022 led to increased system costs.

Sources: Sources: FerroAlloyNet, company disclosures, expert estimates. Approximate values.

EXHIBITS 8

VRFB vs. Lithium-Ion: Levelized Cost Comparison by Duration (2h, 4h, 8h, 10h)

VRFB becomes cost-competitive at 4 hours and clearly advantageous at 8+ hours.



Lithium-ion LCOS increases slightly with duration due to the need for multiple battery packs. VRFB LCOS declines with duration due to the scalability of the electrolyte tank.

Sources: Sources: Lazard, NREL, company data. Assumes 20-year life, 8% discount rate. Values are approximate.

This report was informed by public research and data from: Guidehouseinsights, Bnef, Usgs, Cnesa, Lazard, Ferroalloy.net, Invinity, Sumitomoelectric.
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